PromoFinder

A tool that reads a microdrama's screenplays and hands an editor a ranked list of 20-60 second promos, with exact cut points.

Live app: kuku-promo-finder.streamlit.app · Code: github.com/vatsal3512/kuku-promo-finder · Raw outputs: raw_outputs/ in the repo

Why I picked this one

Right now an editor watches every episode to find promo moments. For a 50-episode show that's about 100 minutes of footage, and Kuku puts out new shows every week. The output is easy to check (timestamps either work or they don't), the brief says clearly what a good promo is, and promos are how paid viewers arrive, so a better pick is worth real money. I also liked that the editor doesn't go away. They still approve every cut. They just start from a shortlist instead of raw footage.

What I assumed

- The timestamps in the scripts match the finished video, and the CAMERA lines mark real shot changes, so I cut on those.
- A promo's job is to bring in people who have never seen the show. So I favour clips from early episodes, where a new viewer lands near the start. Big late twists are still listed, but marked for retargeting people who are already watching.
- "Hook in the first 3 seconds" means the hook line or action starts within 3 s of the clip's start. "Ends on a cliffhanger" means the cut comes within 1.5 s of that line ending, and never in the middle of a line.
- "Makes sense on its own" is checked by a model that sees only the clip, never the rest of the show.
- Violence, gore, coercion and children in danger get flagged for a person to review. Nothing is silently removed.
- Free tiers only, and every model is a single config value, so a stronger one can be swapped in.

How it works

The main design rule: the language model never writes a timestamp. It points at lines in the script, and code works out the cuts. I tested this the hard way. When I asked the same model for promo timestamps directly, 23 of 100 were unusable, and 22 of those stopped at the *start* of the cliffhanger line, cutting off the very moment the promo exists for.

Step	What happens	Done by
Read	Splits each script into scenes, timed lines, speakers and shots. A `check` command shows how a new show parses before any model is called.	code
Spot	Rules score every line as a possible hook or cliffhanger. One model call per episode names its best hook and cliffhanger lines.	rules, model
Build	About 3,000 possible clips per show: single stretches, clips that run across an episode break, and two-part montages. Anything outside 20-60 s, with a late hook or a sloppy end, is thrown out.	code
Judge	About 140 of the best clips, including alternative cuts of strong moments, go to a "cold viewer" model. It scores hook, cliffhanger and standalone, and flags spoilers and ad risks. Every clip is judged twice, and a clip below 3/5 on standalone in either pass is out.	model
Second opinion	A model from a different family re-judges the 25-clip shortlist and can veto a clip.	model
Craft checks	Small penalties for things the judges miss because they read the whole clip at once: no spoken line in the first 3 s, a one-word opening, a long silent tail, length over 50 s.	code
Rank	At most one promo per scene. Pairwise model comparisons set the order. The first one to post (the "control") must come from the first 20% of episodes and carry the fewest ad flags.	code, model

Each promo comes with its segments, hook, cliffhanger, why it works, why it has that rank, the episode it sends viewers to, and any ad flags. Each run also writes an edit list and an ffmpeg cut script. A Streamlit app (live link above) shows each promo with its timeline and transcript, and lets an editor approve or reject it.

Scale and cost

One 50-episode show takes about 140 model calls and 0.45M tokens, roughly 20-25 minutes on NVIDIA's free tier. At normal paid rates of $1-3 per million tokens, that's under $1.50 a show. The rules-only mode runs in 5 seconds for nothing. Only the tagging step grows with the number of episodes; judging is a fixed budget of about 140 clips per show. Answers are cached, so reruns are nearly free. Shows can run side by side. The real limit is the provider's rate limits, so the tool rotates API keys and backs off when throttled. The editor reads about 12 minutes of shortlisted clips instead of watching 100 minutes of footage, and their approvals are logged for tuning later.

How I know it's good

Check	What I found

Are the cuts usable?	15 of 15 in every run. Asking the model for timestamps directly gave 77 of 100.
Do outside judges agree?	Two models that never picked anything scored the lists blind: Mistral Large 4, and MiniMax M3.1, which wasn't part of any run. Compared with the rules-only list, the final list had better hooks (4.53 vs 4.33) and cliffhangers (4.67 vs 4.47). All of them passed the standalone test, against 87% for rules only (Mistral: 100% vs 93%).
Did self-grading fool me?	Yes, once. A mid-day "improvement" looked better to the model that picked the clips (0.76 to 0.85) but worse to Mistral (standalone pass fell from 100% to 80%). That's why the second-opinion step exists. With it, the list went back to the best level MiniMax has seen (0.80). The craft checks didn't move that score, which makes sense: the judges can't see what the checks fix.
What does a person think?	I compared 8 pairs blind: the tool's cut vs the rules-only cut, often of the same moment. I preferred the tool's in 5 of 8. That's one person, so it's a hint, not proof. But my three "no" votes pointed to real problems, which became the craft checks.
Is it stable?	Run twice with the same settings, 9 of the top 10 stayed the same. Settings changes are riskier. When my first version of the craft checks also changed which clips reached the judge, 7 of the 15 promos changed, purely from judge noise. Now the checks run after judging, and the same change moves only 3 cuts, all in the direction intended.

The top of the final list

#	Clip	Opens on / cuts on	Why it's here
1	Ep1 00:39-01:30 52 s	“Because he is marrying me.” Ends as her father refuses to look at her.	The whole premise in one clip: she's jilted at the altar for her own sister. Nothing for ad review to worry about, and viewers land on episode 1. Post this first.
2	Ep5 00:54-01:45 52 s	Her mother's book thrown into the rain. Ends on the stranger's “Come.”	Works with the sound off, and it's the first time the scarred stranger steps in for her.
3	Ep12 00:26-01:09 43 s	“You planted it!” Ends on “Fail, or refuse, and your husband hangs from the gallows.”	The sharpest choice in the show. Digging bare-handed through thorns needs an ad-policy check.
4	Ep3 01:24-02:15 52 s	A stranger floors the guard. Ends as he kneels with a ring in front of the whole church.	The only early clip that shows the soldier from the title, and it ends on “will she say yes?”. There's a punch, so check ad policy.
5	Ep11 00:16-01:15 58 s	“Your husband stole my gold pocket watch!” Ends mid-search of their home.	Easy for a stranger to follow, but long at 58 s. Worth testing against the shorter ones.

Where it picks badly

It reads; viewers watch. All three AI judges gave hooks like a bare "No." or a silent hand on a bedcover (first word at 0:09) four or five out of five, because they could see the tension coming later in the clip. The craft checks now catch the measurable cases. Whether a shot actually stops a thumb can only be judged from the video.

The judges are generous and noisy. Most scores are 4 or 5. The same clip can move by a point between batches, and one clip got a 5 from one judge and a 3 from another. Once my changes were moving scores by less than that, I stopped tuning against the judges.

It repeats storylines. The one-per-scene rule stops the same scene appearing twice, but four of the fifteen promos still came from the poison storyline.

Stage directions give things away. A line like "she gasps, realizing her secret identity has been laid bare" tells the judge something a viewer never sees, so some clips look clearer on paper than they would on screen.

Names drift. The scripts spell the estate six ways and reuse one speaker label for two people; the tool fixes most of it. The real fix for most of these problems is the video: check the first 3 seconds, then use hook rate data once the ads run.

AI tools I used, and where they got it wrong

I used Claude Code as a coding assistant, to write and debug code and run the experiments I asked for. On NVIDIA's free tier, Nemotron 3 Ultra picks and judges, Mistral Large 4 gives the second opinion, and MiniMax M3.1 audits the results at the end. The calls that shaped the tool were mine. I kept it on free NVIDIA models. I held back the doc until the quality numbers improved. I did the blind A/B test myself, and my three "no" votes (an opening that answers a question nobody heard, a one-word opening, a 58 s clip) became the craft checks. And I pushed for a prototype that looks and reads like a real Kuku tool.

The clearest mistake: the model that picked the clips also graded them, and it said my changed version was better when it was worse. Only judges from other model families caught the drop in standalone clarity. I don't trust a picker to grade its own work any more. Smaller misses: the first models suggested had been retired (HTTP 410), and the pairwise step kept putting a violent scene first, which is why the "control" rule exists.